

FINANCE MASTER BIBLE

Every financial concept, formula, strategy, and principle

From basics to advanced — trading, investing, markets, risk, valuation

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Key Formulas Reference

Every important formula in one place

01 — Financial Markets & Asset Classes

What is a Financial Market?

A financial market is any place — physical or digital — where buyers and sellers trade financial instruments. Markets exist to allocate capital efficiently, provide price discovery, and allow risk transfer between participants.

Asset Classes

Asset Class	What it is	Return source	Risk level	Liquidity
Equities (Stocks)	Ownership in companies	Price appreciation + dividends	Medium-High	High
Fixed Income (Bonds)	Loans to governments/companies	Interest payments (coupon)	Low-Medium	High
Cash & Equivalents	T-bills, money market	Interest rate	Very Low	Highest
Real Estate	Property or REITs	Rent + appreciation	Medium	Low-Medium
Commodities	Gold, oil, wheat, copper	Price appreciation	High	Medium
Forex (FX)	Currency exchange rates	Rate differentials	Medium-High	Highest
Cryptocurrencies	Digital assets on blockchain	Price speculation	Very High	High
Derivatives	Contracts derived from assets	Leverage on underlying	Very High	Medium-High
Private Equity	Stakes in private companies	Exit returns	High	Very Low
Hedge Funds	Pooled alternative strategies	Absolute returns	Varies	Low

Market Types

Primary Market	Where new securities are issued for the first time (IPOs, bond issuance). Company raises capital directly.
Secondary Market	Where existing securities are traded between investors. NYSE, NASDAQ, BSE, NSE.
OTC Market	Over-the-counter — direct trading between parties, not on an exchange. Most forex and bonds trade OTC.
Derivatives Mkt	Exchange-traded (CME, NSE F&O;) and OTC (swaps, forwards). Contracts on underlying assets.
Money Market	Short-term debt instruments. T-bills, commercial paper, repos. Maturity under 1 year.
Capital Market	Long-term securities — stocks and bonds with maturity over 1 year.

Major Global Exchanges

Exchange	Country	Index	Market Cap (approx)
NYSE	USA	Dow Jones, S&P; 500	\$25 trillion+
NASDAQ	USA	NASDAQ Composite	\$20 trillion+

02 — Core Financial Concepts

Time Value of Money (TVM)

The most fundamental concept in finance: a rupee today is worth more than a rupee tomorrow because money today can be invested to earn returns. All of finance — valuation, bonds, mortgages, retirement planning — is built on this idea.

Future Value: $FV = PV \times (1 + r)^n$

Present Value: $PV = FV / (1 + r)^n$

Annuity PV: $PV = PMT \times [1 - (1+r)^{-n}] / r$

Perpetuity PV: $PV = PMT / r$

PV	Present Value — what future money is worth today
FV	Future Value — what today's money grows to
r	Interest rate per period
n	Number of periods
PMT	Payment per period (for annuities)

Compounding

Compounding is earning returns on your returns. The more frequently you compound, the faster your money grows. Albert Einstein reportedly called it the 8th wonder of the world.

Compound Interest: $A = P \times (1 + r/n)^{(n \times t)}$

Continuous Compound: $A = P \times e^{(r \times t)}$

Rule of 72: $\text{Years to double} = 72 / \text{annual_rate_percent}$

Rule of 72 examples:

Annual Return	Years to Double	Example
3%	24 years	Savings account
6%	12 years	Conservative portfolio
8%	9 years	Long-term stock average
12%	6 years	Growth portfolio
24%	3 years	Exceptional performance
72%	1 year	Crypto bull run (rare)

Interest Rate Types

Nominal Rate	The stated rate, not adjusted for inflation or compounding frequency.
Real Rate	Nominal rate minus inflation. Real Rate $\approx$ Nominal Rate - Inflation (Fisher equation).
Effective Rate	The actual annual return after compounding. $EAR = (1 + r/n)^n - 1$
Risk-Free Rate	Return on zero-risk investment. US 10-yr Treasury or T-bill used as benchmark.
Discount Rate	Rate used to calculate present value of future cash flows in DCF valuation.
Hurdle Rate	Minimum return required for an investment. Projects below this are rejected.
WACC	Weighted Average Cost of Capital. Blend of equity and debt cost. Used in DCF.

Inflation

Inflation is the rate at which purchasing power decreases over time — prices rise, so each unit of currency buys less. Central banks target ~2% inflation as healthy.

Real Return = Nominal Return - Inflation Rate

Fisher Equation: $(1 + \text{nominal}) = (1 + \text{real}) \times (1 + \text{inflation})$

CPI	Consumer Price Index — measures inflation for typical household basket of goods.
PPI	Producer Price Index — measures inflation at the wholesale/production level.
Core CPI	CPI excluding food and energy (more volatile items). Fed's preferred measure.
PCE	Personal Consumption Expenditures — Fed's actual preferred inflation gauge.
Hyperinflation	Extremely rapid inflation (>50%/month). Destroys currency value. Zimbabwe 2008.
Deflation	Negative inflation — prices fall. Sounds good but causes economic slowdown.
Stagflation	High inflation + low growth + high unemployment. 1970s USA. Very hard to fix.

03 — Technical Analysis

Technical analysis studies price and volume data to forecast future price movements. It assumes all information is already reflected in the price and history repeats itself.

Candlestick Charts

Open	Price at which the period began trading.
High	Highest price reached during the period.
Low	Lowest price reached during the period.
Close	Price at which the period ended. Most important price.
Body	Rectangle between open and close. Green/white = close > open. Red/black = close < open.
Wick/Shadow	Lines above/below body showing high and low extremes.
Volume	Number of shares/units traded. Confirms price moves.

Key Candlestick Patterns

Pattern	Type	Signal	Reliability
Doji	Reversal	Indecision — open = close. Watch context.	Medium
Hammer	Bullish	Long lower wick. Buyers rejected low prices.	Good
Shooting Star	Bearish	Long upper wick. Sellers rejected high prices.	Good
Engulfing Bull	Bullish	Green candle fully engulfs prior red candle.	High
Engulfing Bear	Bearish	Red candle fully engulfs prior green candle.	High
Morning Star	Bullish	3-candle reversal at bottom: down, doji, up.	High
Evening Star	Bearish	3-candle reversal at top: up, doji, down.	High
Marubozu	Trend	No wicks — strong directional conviction.	High
Spinning Top	Neutral	Small body, both wicks. Indecision.	Low
3 White Soldiers	Bullish	3 consecutive strong green candles.	High
3 Black Crows	Bearish	3 consecutive strong red candles.	High

Moving Averages

SMA (Simple)	Equal weight to all periods. SMA-200 = average of last 200 closes. Slow to react.
EMA (Exponential)	More weight to recent prices. Faster than SMA. EMA-12 and EMA-26 used in MACD.
WMA (Weighted)	Linear weighting — most recent gets highest weight. Between SMA and EMA.
VWAP	Volume Weighted Average Price. Used by institutions as benchmark. Intraday only.
Golden Cross	SMA-50 crosses above SMA-200. Strong bullish signal.
Death Cross	SMA-50 crosses below SMA-200. Strong bearish signal.
Ribbon	Multiple EMAs (8,13,21,34,55,89). Spread shows trend strength.

Technical Indicators

Indicator	Type	Formula/Logic	Signal
RSI (14)	Momentum	$100 - 100/(1 + RS)$. RS = avg gain / avg loss	>70 overbought, <30 oversold
MACD	Trend/Momentum	EMA(12) - EMA(26). Signal = EMA(9) of MACD	Crossover above/below signal line
Bollinger Bands	Volatility	SMA $\pm$ 2 standard deviations	Price at band edge = extreme
Stochastic	Momentum	$(\text{Close} - \text{Low}) / (\text{High} - \text{Low}) \times 100$	>80 overbought, <20 oversold
ATR	Volatility	Average True Range over N periods	Higher ATR = more volatile
OBV	Volume	Cumulative volume based on price direction	Divergence signals reversal
ADX	Trend strength	Directional movement index. 0-100	>25 = strong trend
Ichimoku	Multi-signal	5 lines: tenkan, kijun, senkou A/B, chikou	Cloud = support/resistance
CCI	Momentum	$(\text{Price} - \text{SMA}) / (0.015 \times \text{Mean Deviation})$	>100 overbought, <-100 oversold
Williams %R	Momentum	Similar to stochastic, inverted scale	>-20 overbought, <-80 oversold

Chart Patterns

Pattern	Type	Description	Target
Head & Shoulders	Bearish reversal	3 peaks, middle highest. Neckline break = sell.	Depth of head
Inverse H&S;	Bullish reversal	3 troughs, middle lowest. Neckline break = buy.	Depth of head
Double Top	Bearish reversal	Two equal highs at resistance. W shape.	Distance from top to neckline
Double Bottom	Bullish reversal	Two equal lows at support. M shape.	Distance bottom to neckline

Ascending Triangle	Bullish	Flat resistance, rising support. Breakout up.	Triangle height
Descending Triangle	Bearish	Flat support, falling resistance. Breakdown.	Triangle height
Symmetrical Triangle	Neutral	Converging highs and lows. Breakout either way.	Triangle height
Cup & Handle	Bullish	Rounded bottom like a cup, small pullback handle.	Cup depth
Bull Flag	Bullish	Sharp rise (pole), tight consolidation (flag).	Pole height
Bear Flag	Bearish	Sharp drop (pole), tight bounce (flag).	Pole height
Wedge Rising	Bearish	Both lines rising but converging. Breakdown likely.	Entry point
Wedge Falling	Bullish	Both lines falling but converging. Breakout likely.	Entry point

Support & Resistance

Support	Price level where buying interest is strong enough to stop the price falling. Demand zone.
Resistance	Price level where selling interest stops the price rising. Supply zone.
Role reversal	Once broken, support becomes resistance and vice versa. Key concept.
S/R levels	Identified from previous highs/lows, round numbers, VWAP, Fibonacci levels.
Fibonacci	Key retracement levels: 23.6%, 38.2%, 50%, 61.8% (golden ratio), 78.6%

04 — Fundamental Analysis

Fundamental analysis evaluates a company's intrinsic value by examining financial statements, industry position, management quality, and macroeconomic factors. The goal is to find stocks trading below their true worth.

Key Valuation Ratios

Ratio	Formula	What it means	Good range
P/E Ratio	Price / EPS	How much you pay per rupee of earnings	15-25x typical; <15 cheap
P/B Ratio	Price / Book Value per share	Price vs accounting value of company	<1 undervalued; >3 expensive
EV/EBITDA	Enterprise Value / EBITDA	Better P/E for capital-intensive firms	5-10x typical
P/S Ratio	Price / Revenue per share	Used for unprofitable growth companies	<2 cheap; >10 expensive
PEG Ratio	P/E / Earnings Growth Rate	Adjusts P/E for growth. <1 = undervalued	<1 attractive
Dividend Yield	Annual Dividend / Price	Income return from dividends	2-5% typical
ROE	Net Income / Equity	How efficiently equity generates profit	>15% good
ROA	Net Income / Total Assets	Profit per asset dollar	>5% good
ROCE	EBIT / Capital Employed	Return on all capital used	>15% good
Debt/Equity	Total Debt / Equity	Financial leverage	<1 conservative; >2 risky
Current Ratio	Current Assets / Current Liabilities	Short-term liquidity	>1.5 safe
Quick Ratio	(Current Assets - Inventory) / Current Liabilities	Strict liquidity test	>1 safe
Interest Coverage	EBIT / Interest Expense	Can company pay its interest?	>3x safe
Asset Turnover	Revenue / Total Assets	How well assets generate revenue	Higher = better
Gross Margin	Gross Profit / Revenue	Core business profitability	Industry dependent
Net Margin	Net Income / Revenue	Bottom line profitability	>10% good for most
EPS	Net Income / Shares Outstanding	Profit per share	Growing YoY = good
Free Cash Flow Yield	FCF / Market Cap	Cash generation vs price	>5% attractive

Warren Buffett's Investing Checklist

- Competitive moat — durable competitive advantage (brand, network effect, switching cost, cost advantage)
- Consistent earnings growth over 10+ years
- Low debt — able to pay off total debt within 3-4 years from earnings
- High and stable Return on Equity (ROE > 15%) without excessive leverage
- Management that allocates capital intelligently (buybacks, dividends, acquisitions)
- Simple business that is easy to understand (circle of competence)
- Price with margin of safety — buy at significant discount to intrinsic value

- Long-term perspective — hold forever if business remains strong

Industry Analysis — Porter's Five Forces

Competitive Rivalry	How intense is competition between existing players? More = worse margins.
Threat of New Entrants	How easy is it for new companies to enter? Low barriers = bad for incumbents.
Bargaining Power (Buyers)	Can customers push prices down? Commodity products = high buyer power.
Bargaining Power (Suppliers)	Can suppliers raise prices? Few suppliers = bad for company.
Threat of Substitutes	Can customers switch to alternatives? High threat = price ceiling.

05 — Portfolio Theory & Diversification

Modern Portfolio Theory (MPT) — Harry Markowitz

MPT (1952) shows that investors can construct portfolios to maximize expected return for a given level of risk. The key insight: combining assets that don't move together (low correlation) reduces total portfolio risk without reducing returns.

Portfolio Return: $E(R_p) = \sum w_i \times E(R_i)$

Portfolio Variance: $\sigma_p^2 = \sum \sum w_i \times w_j \times \sigma_i \times \sigma_j \times \rho_{ij}$

Portfolio Std Dev: $\sigma_p = \sqrt{\sigma_p^2}$

Correlation (ρ)	-1 = perfectly inverse, 0 = uncorrelated, +1 = perfectly together. Want low ρ .
Efficient Frontier	Set of optimal portfolios offering max return for each risk level. Curve on risk-return chart.
Minimum Variance	Portfolio with lowest possible risk. Often holds bonds + stocks.
Sharpe Optimal	Tangency portfolio on efficient frontier — highest Sharpe ratio point.
Diversification	Spreading across assets reduces unsystematic risk. 20-30 stocks eliminates most idiosyncratic risk.

Types of Risk

Systematic Risk	Market risk. Cannot be diversified away. Beta measures it. Examples: recession, war.
Unsystematic Risk	Company/sector specific. Can be eliminated through diversification.
Market Risk	Risk of overall market declining. Measured by Beta.
Credit Risk	Risk that borrower defaults on debt obligation.
Liquidity Risk	Risk of not being able to sell an asset quickly at fair price.
Inflation Risk	Risk that inflation erodes real returns. Bonds most exposed.
Currency Risk	Risk of exchange rate movements when holding foreign assets.
Concentration Risk	Risk of too much exposure to one stock, sector, or geography.
Tail Risk	Risk of extreme, unlikely events (black swans). Often underestimated.
Counterparty Risk	Risk that the other party in a transaction defaults.

Asset Allocation Frameworks

Framework	Allocation	Best for
60/40 Classic	60% stocks, 40% bonds	Moderate risk, balanced growth
All Weather (Dalio)	30% stocks, 40% LT bonds, 15% IT bonds, 7.5% gold, 7.5% commodities	Survives all economic environments

Permanent Portfolio	25% stocks, 25% bonds, 25% gold, 25% cash	Ultra conservative, stable
100-Age Rule	Stocks % = 100 - your age	Reduces risk as you get older
3-Fund Portfolio	US stocks, International stocks, Bonds	Simple, low cost, diversified
Warren Buffett's	90% S&P; 500 index, 10% T-bills (for wife)	Simple long-term growth
Aggressive Growth	90%+ equities, small/mid cap tilt	Young investors, long horizon

CAPM — Capital Asset Pricing Model

CAPM describes the relationship between systematic risk (beta) and expected return. It says you should only be compensated for market risk, not for risks you could diversify away.

$$\text{CAPM: } E(R_i) = R_f + \beta_i \times (E(R_m) - R_f)$$

$$\text{Beta: } \beta = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$$

R_f Risk-free rate (T-bill rate)

E(R_m) Expected market return (historical ~8-10% for S&P; 500)

E(R_m) - R_f Equity risk premium (ERP). Typically 5-7%

β = 1 Moves with the market exactly

β > 1 More volatile than market. Tech stocks often β = 1.5-2

β < 1 Less volatile. Utilities, consumer staples β = 0.3-0.7

β = 0 No correlation with market. Gold is close to this

β < 0 Inverse to market. Rare. Gold miners sometimes negative beta

Key Risk Metrics

Metric	Formula	Meaning	Good value
Sharpe Ratio	$(\text{Return} - R_f) / \text{Std Dev}$	Return per unit of total risk	>1 good, >2 excellent
Sortino Ratio	$(\text{Return} - R_f) / \text{Downside Dev}$	Only penalizes downside risk	>2 good
Calmar Ratio	$\text{Ann. Return} / \text{Max Drawdown}$	Return per unit of worst drawdown	>2 good
Information Ratio	$\text{Active Return} / \text{Tracking Error}$	Consistency of outperformance vs benchmark	>0.5 good
Treynor Ratio	$(\text{Return} - R_f) / \text{Beta}$	Return per unit of market risk	Higher = better
Max Drawdown	$(\text{Peak} - \text{Trough}) / \text{Peak}$	Worst peak-to-trough decline	<20% conservative
VaR (95%)	5th percentile of return distribution	Worst loss expected 95% of the time	Depends on strategy
CVaR / ES	Average loss beyond VaR	Expected loss in worst scenarios	Depends on strategy
Win Rate	$\text{Winning trades} / \text{Total trades}$	% of profitable trades	>50% desirable
Profit Factor	$\text{Gross Profit} / \text{Gross Loss}$	Total wins vs total losses	>1.5 good
Avg Win/Loss	$\text{Avg winning trade} / \text{Avg losing trade}$	Risk-reward per trade	>1.5 good

Position Sizing Methods

Fixed %	Risk fixed % of portfolio per trade. Common: 1-2%. Simple and protects capital.
Kelly Criterion	$f = (bp - q) / b$. Optimal bet size. Often use half-Kelly for safety.
ATR-Based	Size inversely to ATR. More volatile = smaller position. Keeps \$ risk constant.
Volatility Target	Target a fixed portfolio volatility (10-15%). Size all positions to achieve it.
Equal Weight	Same \$ amount in each position. Simple, diversified.
Risk Parity	Weight positions so each contributes equally to total portfolio risk.

Kelly Criterion: $f^* = (bp - q) / b$

where: b = odds (profit/loss ratio), p = win probability, $q = 1 - p$

ATR Position Size: $\text{Shares} = (\text{Portfolio} \times \text{Risk\%}) / (\text{ATR} \times \text{Multiplier})$

Stop Loss Strategies

Fixed % Stop	Exit if price falls X% from entry. Simple but ignores volatility.
ATR Stop	$\text{Stop} = \text{Entry} - (2 \times \text{ATR})$. Adapts to volatility. Professional standard.

Support Stop	Stop just below key support level. Logical placement.
Time Stop	Exit if trade hasn't moved in your favor within N days.
Trailing Stop	Stop moves up with price. Locks in profits while allowing upside.
Chandelier Stop	Highest close - (3 × ATR). Popular trailing stop method.
Mental Stop	No hard stop, rely on judgment. Dangerous — emotions override discipline.

Drawdown Analysis

$\text{Max Drawdown} = (\text{Peak Value} - \text{Trough Value}) / \text{Peak Value}$

$\text{Drawdown Duration} = \text{Days from peak to new peak (recovery time)}$

$\text{Underwater Curve} = \text{Cumulative drawdown over time}$

Peak-to-Trough	The single largest decline from any high point to the subsequent low.
Recovery Time	Days taken to recover from a drawdown back to previous peak. Critical metric.
Underwater Period	Time spent below previous peak. Long periods = strategy is struggling.
Calmar trade-off	Higher Calmar = better risk-adjusted return. Target Calmar > 2.0.

Strategy Categories

Strategy	Concept	Works in	Timeframe	Key risk
Trend Following	Buy strength, sell weakness. Ride momentum.	Trending markets	Days to months	Whipsaws in ranging markets
Mean Reversion	Buy when oversold, sell when overbought.	Ranging markets	Hours to days	Trend continuation
Momentum	Rank assets, buy top performers, short bottom.	Trending markets	Monthly rebalance	Momentum crash
Carry Trade	Borrow in low-rate currency, invest in high-rate.	Low volatility	Weeks to months	Sudden reversal
Arbitrage	Exploit price differences of same asset.	Always	Milliseconds	Execution risk
Statistical Arb	Pairs trading based on statistical relationships.	Always	Hours to days	Correlation breakdown
Event-Driven	Trade around earnings, mergers, spin-offs.	Specific events	Days to weeks	Binary outcomes
Breakout	Buy when price breaks above resistance.	Trending	Days to weeks	False breakouts
Value Investing	Buy undervalued stocks, hold for revaluation.	All markets	Years	Value trap
Growth Investing	Buy high-growth companies at reasonable price.	Bull markets	Years	High multiples contract
Volatility Trading	Trade implied vs realized volatility.	All markets	Days to weeks	Vol spikes

Dual Momentum Strategy (Gary Antonacci)

Combines absolute momentum (trending above zero) with relative momentum (ranking). One of the most backtested, simple, robust strategies available.

Step 1	Calculate 12-month return for each asset in universe.
Step 2	Check absolute momentum: is the return positive? (Beat T-bills?)
Step 3	Rank by relative momentum: which performed best?
Step 4	Buy the top-ranked asset IF it has positive absolute momentum.
Step 5	If no asset has positive momentum, go to cash/bonds.
Rebalance	Monthly. Hold for 1 month minimum.
Universe	Can use just SPY + international equity + bonds for simplicity.

Historical

~12% CAGR, <15% max drawdown over 40+ years. Calmar ~0.8-1.2.

Mean Reversion — Key Concepts

Z-Score	How many standard deviations from mean. $ Z > 2$ = extreme, likely to revert.
Bollinger Bands	Price outside bands (2 SD) = overbought/oversold. High probability reversion.
RSI Reversal	RSI < 30 (oversold) = buy signal. RSI > 70 (overbought) = sell signal.
Pairs Trading	Long underperformer, short outperformer of correlated pair. Profit from convergence.
Half-Life	How long it takes for a spread to revert halfway. Shorter = more tradeable.

Risk-Reward and Trade Management

$\text{Risk-Reward Ratio} = \text{Potential Profit} / \text{Potential Loss}$

Minimum R:R for 50% win rate to be profitable = 1:1 (break even at 1.5:1 after costs)

$\text{Expected Value} = (\text{Win Rate} \times \text{Avg Win}) - (\text{Loss Rate} \times \text{Avg Loss})$

For long-term profitability you need positive expected value:

Win Rate	Min R:R needed	Example
30%	2.3:1	High R:R trend systems
40%	1.5:1	Most swing trading
50%	1.0:1	Break even (need >1 for profit)
60%	0.67:1	Mean reversion systems
70%	0.43:1	High win rate scalping

Options Basics

Call Option	Right (not obligation) to BUY the underlying at the strike price before expiry.
Put Option	Right (not obligation) to SELL the underlying at the strike price before expiry.
Strike Price	The agreed price at which you can buy/sell the underlying.
Expiration	Date when the option expires worthless if not exercised.
Premium	Price paid to buy the option. Maximum loss for buyer.
In The Money	Call: underlying > strike. Put: underlying < strike. Has intrinsic value.
At The Money	Strike $\approx$ underlying price. No intrinsic value, only time value.
Out of Money	Call: underlying < strike. Put: underlying > strike. Only time value.
Intrinsic Value	Value if exercised immediately. Call: $\max(S-K, 0)$. Put: $\max(K-S, 0)$.
Time Value	Premium - Intrinsic Value. Decays to zero at expiry.
American Style	Can be exercised at any time before expiry.
European Style	Can only be exercised at expiry. Most index options.

The Greeks

The Greeks measure how option price changes with various factors. Essential for hedging.

Greek	Symbol	Measures	Call value	Put value
Delta	Δ	Change in option price per \$1 change in underlying	0 to +1	-1 to 0
Gamma	Γ	Rate of change of Delta. Highest at ATM	Always positive	Always positive
Theta	Θ	Time decay — option loses value each day	Negative (buyer loses)	Negative (buyer loses)
Vega	v	Sensitivity to implied volatility change	Positive (long vol)	Positive (long vol)
Rho	ρ	Sensitivity to interest rate change	Positive	Negative

Black-Scholes Model

$$\text{Call Price} = S \times N(d1) - K \times e^{(-rT)} \times N(d2)$$

$$\text{Put Price} = K \times e^{(-rT)} \times N(-d2) - S \times N(-d1)$$

$$d1 = [\ln(S/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T})$$

$$d2 = d1 - \sigma\sqrt{T}$$

S	Current stock price
K	Strike price
r	Risk-free interest rate
T	Time to expiry (in years)
σ	Implied volatility (annual)
N()	Cumulative normal distribution function

Common Options Strategies

Strategy	Construction	Outlook	Max Profit	Max Loss
Long Call	Buy call	Bullish	Unlimited	Premium paid
Long Put	Buy put	Bearish	Strike - Premium	Premium paid
Covered Call	Own stock + sell call	Neutral/slight bullish	Strike - Stock + Premium	Stock price - Premium
Protective Put	Own stock + buy put	Bullish with protection	Unlimited	Stock fall - Premium
Bull Call Spread	Buy low strike call, sell high strike call	Moderately bullish	Spread - Premium	Net premium
Bear Put Spread	Buy high strike put, sell low strike put	Moderately bearish	Spread - Premium	Net premium
Straddle	Buy call + put, same strike	Expecting big move, unknown direction	Unlimited	Both premiums
Strangle	Buy OTM call + OTM put	Expecting big move (cheaper than straddle)	Unlimited	Both premiums
Iron Condor	Sell strangle + buy wider strangle	Range-bound, low volatility	Net premium	Spread - Premium
Calendar Spread	Sell near-term, buy far-term same strike	Low volatility near-term	Limited	Net debit

Futures & Forwards

Futures	Standardized exchange-traded contracts to buy/sell at future date. Daily mark-to-market.
Forwards	OTC customized contracts. No daily settlement. Counterparty risk.
Basis	Spot price - Futures price. Converges to zero at expiry.
Contango	Futures price > spot price. Normal for commodities (storage costs).
Backwardation	Futures price < spot price. Happens when immediate supply is tight.
Margin	Good faith deposit, not full price. Provides leverage. ~5-10% of contract value.

Bond Fundamentals

Face Value/Par	The amount returned to bondholder at maturity. Typically \$1,000 or Rs.1,000.
Coupon Rate	Annual interest rate on face value. 6% coupon on \$1,000 = \$60/year.
Maturity	Date when principal is repaid. Short (<2yr), medium (2-10yr), long (>10yr).
Price	Bond trades at premium (>100), par (100), or discount (<100) based on rates.
Yield	Effective return given current price. Inverse relationship with price.
YTM	Yield to Maturity — total return if held to maturity. Most important yield measure.
Current Yield	Annual Coupon / Current Price. Simple approximation.
YTM formula	Requires solving: $\text{Price} = \sum [\text{Coupon}/(1+\text{YTM})^t] + [\text{Face}/(1+\text{YTM})^n]$

Bond Price and Interest Rate Relationship

Bond prices and interest rates move INVERSELY. If rates go up, existing bond prices fall because new bonds offer better rates. This is the fundamental risk in bond investing.

$$\text{Bond Price} = \sum [C/(1+r)^t] + [F/(1+r)^n]$$

$$\text{Duration: } \Delta P/P \approx -\text{Duration} \times \Delta r$$

$$\text{Modified Duration} = \text{Macaulay Duration} / (1 + \text{YTM}/n)$$

Duration	Weighted average time to receive all cash flows. Sensitivity measure.
Modified Duration	% price change for 1% change in yield. 5yr MD = -5% for +1% rate rise.
Convexity	Curvature of price-yield relationship. Higher convexity = less price drop when rates rise.
DV01	Dollar Value of 01 — price change for 0.01% (1bp) move in yield.

Yield Curve

Normal Curve	Long-term yields > short-term. Healthy economy, normal expectations.
Inverted Curve	Short-term yields > long-term. Reliable recession predictor. Happened before every US recession.
Flat Curve	Short and long yields similar. Transition period, uncertainty.
Steepening	Spread between long and short widening. Often bullish for economy.
2s10s Spread	10yr Treasury yield minus 2yr yield. Most watched recession indicator.
Fed Funds Rate	Overnight lending rate between banks. Set by Federal Reserve. Controls short end of curve.

Credit Ratings

Rating (S&P;/Moody's)	Category	Meaning	Typical yield spread
AAA / Aaa	Investment Grade	Highest quality, minimal credit risk	+10 to 50 bps
AA / Aa	Investment Grade	Very high quality	+50 to 100 bps
A / A	Investment Grade	Upper medium grade	+100 to 150 bps
BBB / Baa	Investment Grade (Lowest)	Medium grade. Many funds must hold IG	+150 to 250 bps
BB / Ba	High Yield / Junk	Speculative, some risk	+250 to 400 bps
B / B	High Yield	Speculative, significant risk	+400 to 600 bps
CCC / Caa	High Yield	Poor standing, vulnerable to default	+600 to 1000+ bps
D / D	Default	In default or breach of obligation	N/A

10 — Macroeconomics & Market Cycles

Key Economic Indicators

Indicator	What it measures	Market impact	Release frequency
GDP	Total economic output	High GDP = bullish stocks, bearish bonds	Quarterly
CPI	Consumer inflation	High CPI = Fed hikes rates, bearish bonds/growth stocks	Monthly
PCE	Fed's preferred inflation gauge	Same as CPI but more direct Fed impact	Monthly
Non-Farm Payrolls	Job creation ex-agriculture	Strong jobs = economy healthy, may hike rates	First Friday monthly
Unemployment Rate	% of labor force unemployed	Low = strong economy. NAIRU ~4-5%	Monthly
PMI Manufacturing	Factory activity (>50=expansion)	Leading indicator of economy	Monthly
PMI Services	Services activity (>50=expansion)	Services-dominant economies watch this closely	Monthly
Retail Sales	Consumer spending	Consumers = 70% of US economy	Monthly
Housing Starts	New home construction	Leading indicator, rate-sensitive	Monthly
Trade Balance	Exports minus imports	Deficit = foreign demand for currency	Monthly
Consumer Confidence	Survey of household sentiment	Forward-looking spending indicator	Monthly
Yield Curve (2s10s)	10yr - 2yr Treasury spread	Inversion = recession warning within 12-18 months	Daily

Economic Cycles — Ray Dalio's Framework

Dalio identifies two main cycle types that drive markets. Understanding where you are in the cycle helps allocate capital correctly.

Short-Term Cycle	Business cycle. 5-8 years. Expansion → Peak → Contraction → Trough.
Long-Term Cycle	Debt super-cycle. 50-75 years. Built on credit expansion then deleveraging.
Expansion	GDP growing, unemployment falling, corporate earnings rising. Buy: stocks, real estate.
Peak	Economy at full capacity, inflation rising, Fed hiking rates. Reduce risk.
Contraction	GDP slowing/declining, unemployment rising. Buy: bonds, gold, defensive stocks.
Trough	Bottom of recession. Fed cutting rates aggressively. Buy: equities aggressively.
Reflation	Recovery beginning. Buy: commodities, cyclicals, small caps.

Central Banks & Monetary Policy

Federal Reserve	US central bank. Dual mandate: stable prices + maximum employment.
ECB	European Central Bank. Mandate: price stability only (2% inflation).
RBI	Reserve Bank of India. Manages rupee, inflation, banking regulation.
Rate Hikes	Increase borrowing costs → slows economy, reduces inflation → bearish stocks/bonds.
Rate Cuts	Decrease borrowing costs → stimulates economy → bullish stocks, bonds.
QE (Quantitative Easing)	Central bank buys bonds → injects money → lowers long rates → bullish all assets.
QT (Quantitative Tightening)	Selling bonds or letting them mature → removes money → bearish all assets.
Forward Guidance	Communication about future policy intentions. Markets price this in immediately.
Taylor Rule	Formula for setting Fed Funds Rate: $r = r^* + 1.5(\pi - \pi^*) + 0.5(y - y^*)$
Dot Plot	Fed's projection of future rate path. Published quarterly. Closely watched.

Cognitive Biases in Trading

Bias	Description	How it hurts traders
Overconfidence	Overestimate ability and accuracy of predictions	Takes too large positions, ignores risk
Loss Aversion	Losses feel 2x more painful than equivalent gains	Holds losers too long, cuts winners too early
Anchoring	Over-rely on first piece of information seen	Stuck on purchase price, ignores new data
Confirmation Bias	Seek information that confirms existing view	Ignores contradicting signals
Herding	Follow the crowd regardless of fundamentals	Buys tops, sells bottoms with the herd
Recency Bias	Over-weight recent events in decision making	Extrapolates recent trend forever
Gambler's Fallacy	Believe past random events affect future	'It's due for a reversal after 5 red candles'
Hindsight Bias	Believe past events were predictable	Overconfidence in ability to predict future
Availability Bias	Weight easily recalled events too heavily	Over-fear crashes after recent crash news
Disposition Effect	Sell winners too early, hold losers too long	Destroys returns systematically
Status Quo Bias	Prefer doing nothing over making changes	Holds bad positions, misses rebalancing
Framing Effect	Decision changes based on how option is presented	Susceptible to marketing of products
Sunk Cost Fallacy	Continue investment because of past investment	'I can't sell at a loss now'
FOMO	Fear of missing out drives impulsive entry	Buys parabolic moves at the top

Market Anomalies (Exploitable Patterns)

Momentum Effect	Stocks that outperformed over 3-12 months continue outperforming. Strongest anomaly.
Value Premium	Low P/B, P/E stocks outperform long-term. Fama-French factor.
Size Premium	Small cap stocks outperform large caps long-term. Higher risk premium.
Low Volatility	Low volatility stocks outperform high volatility stocks. Contradicts CAPM.
January Effect	Stocks (especially small caps) tend to rise in January. Tax-loss selling reversal.
Sell in May	Stock returns historically lower May-October vs November-April.
Weekend Effect	Returns negative on Mondays historically. Less relevant now.
Earnings Drift	Stocks continue moving in direction of earnings surprise for 60+ days (PEAD).
IPO Underperformance	IPOs underperform the market over 3-5 years on average.
Dogs of the Dow	Buying 10 highest-yield Dow stocks annually has beaten index historically.

Stages of a Market Bubble (Hyman Minsky)

- Displacement — New technology, innovation, or policy shift creates new profit opportunity
- Boom — Early investors profit, prices rise gradually, media coverage begins
- Euphoria — Prices detach from fundamentals, 'this time is different' mentality, everyone buying
- Profit Taking — Smart money starts selling quietly, insiders distribute to retail investors
- Panic — Prices collapse rapidly, forced selling, liquidity evaporates, credit crunch

Backtesting Principles

In-sample	Data used to develop and optimize the strategy. Never report results on this alone.
Out-of-sample	Data the strategy has never seen. True performance test.
Walk-forward	Repeatedly optimize on rolling in-sample, test on subsequent out-of-sample periods.
Overfitting	Strategy performs perfectly on historical data but fails live. Too many parameters.
Lookahead Bias	Using data that wasn't available at decision time. Destroys backtest validity.
Survivorship Bias	Testing only on stocks that survived — ignores delisted companies. Inflates returns.
Transaction Costs	Always include commissions, slippage, market impact. Destroys many strategies.
Slippage	Difference between expected and actual execution price. Especially bad for illiquid.

Quant Strategy Building Process

1. Hypothesis — Start with an economic reason WHY this should work (not data mining)
2. Universe — Define which assets to trade (liquidity filter, market cap filter)
3. Signal — Define entry and exit signals clearly in mathematical terms
4. Portfolio construction — How to size positions, rebalancing frequency
5. Risk management — Stop losses, drawdown limits, leverage constraints
6. Backtest — Test on in-sample data, record metrics honestly
7. Walk-forward test — Validate on out-of-sample data
8. Paper trading — Run live simulation without real money for 1-3 months
9. Live deployment — Start small, scale up gradually after validation
10. Monitor — Track if live performance matches backtest expectations

Key Quant Metrics to Track

Metric	Formula	Target
CAGR	$(\text{End}/\text{Start})^{(1/\text{years})} - 1$	>10% annualized
Max Drawdown	$\text{Max}(\text{Peak}-\text{Trough})/\text{Peak}$	<20% conservative
Calmar Ratio	$\text{CAGR} / \text{Max Drawdown}$	>2.0
Sharpe Ratio	$(\text{Return} - R_f) / \text{Std Dev} \times \sqrt{252}$	>1.0, ideally >1.5
Sortino Ratio	$(\text{Return} - R_f) / \text{Downside Std Dev} \times \sqrt{252}$	>2.0
Win Rate	$\text{Winning trades} / \text{Total trades}$	>45%
Profit Factor	$\text{Gross profit} / \text{Gross loss}$	>1.5
Avg Win/Loss	$\text{Avg winning trade} / \text{Avg losing trade}$	>1.5:1
Correlation to market	Pearson correlation vs benchmark	<0.7 for true alpha
Beta	Strategy beta vs market	<1.0 ideal
Alpha	Excess return beyond CAPM expectation	>0

13 — Financial Statements

Income Statement (P&L;)

Revenue	Total sales / turnover. Top line. Growing revenue = healthy business.
COGS	Cost of Goods Sold. Direct costs of producing revenue.
Gross Profit	Revenue - COGS. Gross Margin = Gross Profit / Revenue.
SG&A;	Selling, General & Administrative expenses. Overhead costs.
R&D;	Research & Development spending. Invested in future products.
EBITDA	Earnings before interest, taxes, depreciation, amortization. Operating performance.
EBIT/Operating Profit	EBITDA - D&A.; Earnings from core operations.
EBT	Earnings before tax. After interest expense.
Net Income	Bottom line. After all expenses and taxes. EPS = Net Income / Shares.
D&A;	Depreciation & Amortization. Non-cash expense reducing taxable income.

Balance Sheet

Assets = Liabilities + Equity (Always balances)	
Current Assets	Cash, receivables, inventory. Convertible to cash within 1 year.
Non-Current Assets	PP&E;, goodwill, intangibles. Long-term assets.
Current Liabilities	Payables, short-term debt, deferred revenue. Due within 1 year.
Long-Term Liabilities	Long-term debt, pension obligations. Due beyond 1 year.
Shareholders Equity	Assets minus Liabilities. What shareholders actually own. Book value.
Retained Earnings	Cumulative profits kept in business, not paid as dividends.
Goodwill	Excess paid over fair value in acquisition. Intangible asset.
Working Capital	Current Assets - Current Liabilities. Day-to-day liquidity buffer.

Cash Flow Statement

The most important statement — cash is king. Profit can be manipulated with accounting. Cash flow cannot. Buffett focuses heavily on free cash flow.

Operating CF (CFO)	Cash from core business operations. Net income + D&A; - working capital changes.
Investing CF (CFI)	Cash from buying/selling assets. CapEx is the main item. Usually negative = good.

Financing CF (CFF)	Cash from debt and equity. Debt raised, dividends paid, stock buybacks.
Free Cash Flow (FCF)	CFO - CapEx. The most important single cash flow metric. FCF = actual earnings power.
FCF Yield	FCF / Market Cap. >5% is attractive. Compare to bond yield.
CapEx	Capital expenditure — investment in property, plant, equipment for future growth.
Maintenance CapEx	CapEx just to maintain existing assets. Required spending.
Growth CapEx	CapEx for new capacity/growth. Optional. Shows management ambition.

Discounted Cash Flow (DCF)

DCF is the theoretical gold standard of valuation. It values a company at the present value of all future free cash flows discounted at WACC.

$$\text{Intrinsic Value} = \sum [\text{FCF}_t / (1+\text{WACC})^t] + \text{Terminal Value} / (1+\text{WACC})^n$$

$$\text{Terminal Value (Gordon Growth)} = \text{FCF}_n \times (1+g) / (\text{WACC} - g)$$

$$\text{WACC} = (E/V \times R_e) + (D/V \times R_d \times (1-\text{Tax}))$$

FCF	Free Cash Flow in each projection period (typically 5-10 years)
WACC	Weighted Average Cost of Capital. Discount rate.
g	Terminal growth rate. Usually 2-3% (GDP growth rate). Never higher.
Re	Cost of equity (CAPM)
Rd	Cost of debt (after-tax)
Sensitivity	DCF is highly sensitive to assumptions. Always run bull/bear/base scenarios.
Weakness	Garbage in, garbage out. Small changes in g or WACC change value dramatically.

Comparable Company Analysis (Comps)

Value a company relative to similar publicly traded peers using market multiples. Fast, market-based, but requires truly comparable peers.

Multiple	Formula	Best for	Industry typical
EV/EBITDA	Enterprise Value / EBITDA	Capital-intensive industries	6-12x
P/E	Market Cap / Net Income	Profitable, stable companies	15-25x
P/S	Market Cap / Revenue	High-growth, pre-profit companies	1-10x
P/B	Market Cap / Book Value	Banks, financial companies	1-3x
EV/Revenue	EV / Revenue	SaaS, growth companies	3-15x
EV/EBIT	EV / EBIT	Comparing different tax structures	10-20x
P/FCF	Market Cap / FCF	Cash-generative businesses	15-30x

Precedent Transactions

Value a company based on prices paid for similar companies in past M&A; deals. Usually higher than trading multiples because of control premium (20-30% typical).

Sum of Parts (SOTP) Valuation

Value each business division separately, then add up. Used for conglomerates. Often reveals 'hidden value' — total is worth more than the combined entity trades at.

15 — Master Formula Reference Sheet

TIME VALUE OF MONEY

$$FV = PV \times (1 + r)^n$$

$$PV = FV / (1 + r)^n$$

$$\text{Annuity PV} = PMT \times [1 - (1+r)^{-n}] / r$$

$$\text{Perpetuity PV} = PMT / r$$

$$\text{Rule of 72: Years to double} = 72 / \text{rate\%}$$

$$EAR = (1 + r/n)^n - 1$$

$$\text{Real Rate} \approx \text{Nominal Rate} - \text{Inflation}$$

PORTFOLIO METRICS

$$\text{Portfolio Return} = \sum (w_i \times R_i)$$

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

$$\text{Sortino Ratio} = (R_p - R_f) / \sigma_{\text{downside}}$$

$$\text{Calmar Ratio} = \text{CAGR} / \text{Max Drawdown}$$

$$\text{Beta} = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$$

$$\text{CAPM: } E(R_i) = R_f + \beta \times (E(R_m) - R_f)$$

$$\text{Jensen's Alpha} = R_p - [R_f + \beta(R_m - R_f)]$$

$$\text{Information Ratio} = \text{Active Return} / \text{Tracking Error}$$

$$\text{Max Drawdown} = (\text{Peak} - \text{Trough}) / \text{Peak}$$

$$\text{VaR (parametric)} = \mu - z \times \sigma$$

EQUITY VALUATION

$EPS = \text{Net Income} / \text{Shares Outstanding}$

$P/E = \text{Price} / EPS$

$PEG = P/E / \text{Earnings Growth Rate}$

$EV = \text{Market Cap} + \text{Debt} - \text{Cash}$

$EV/EBITDA = \text{Enterprise Value} / EBITDA$

$P/B = \text{Price} / \text{Book Value per Share}$

DCF: $V = \sum FCFT / (1+WACC)^t + TV / (1+WACC)^n$

$\text{Terminal Value} = FCF_n(1+g) / (WACC-g)$

$WACC = E/V \times Re + D/V \times Rd \times (1-T)$

Gordon Growth: $P = D1 / (r - g)$

PROFITABILITY RATIOS

$\text{Gross Margin} = \text{Gross Profit} / \text{Revenue}$

$\text{Operating Margin} = EBIT / \text{Revenue}$

$\text{Net Margin} = \text{Net Income} / \text{Revenue}$

$EBITDA \text{ Margin} = EBITDA / \text{Revenue}$

$ROE = \text{Net Income} / \text{Shareholders Equity}$

$ROA = \text{Net Income} / \text{Total Assets}$

$ROCE = EBIT / \text{Capital Employed}$

$FCF = CFO - CapEx$

$FCF \text{ Yield} = FCF / \text{Market Cap}$

$\text{Dividend Yield} = DPS / \text{Price}$

TECHNICAL ANALYSIS

$$\text{SMA}(n) = (P_1 + P_2 + \dots + P_n) / n$$

$$\text{EMA} = \text{Price} \times k + \text{EMA}_{\text{prev}} \times (1-k), \quad k = 2/(n+1)$$

$$\text{RSI} = 100 - [100 / (1 + \text{RS})], \quad \text{RS} = \text{Avg Gain} / \text{Avg Loss}$$

$$\text{MACD} = \text{EMA}(12) - \text{EMA}(26)$$

$$\text{Signal Line} = \text{EMA}(9) \text{ of MACD}$$

$$\text{Bollinger Upper} = \text{SMA}(20) + 2\sigma$$

$$\text{Bollinger Lower} = \text{SMA}(20) - 2\sigma$$

$$\text{ATR} = \max(\text{High-Low}, |\text{High-PrevClose}|, |\text{Low-PrevClose}|)$$

$$\text{Stochastic \%K} = (\text{Close-Low}_n) / (\text{High}_n - \text{Low}_n) \times 100$$

$$\text{OBV} = \text{OBV}_{\text{prev}} + \text{Volume (if close up)} \text{ or } - \text{Volume (if close down)}$$

FIXED INCOME

$$\text{Bond Price} = \sum C/(1+r)^t + F/(1+r)^n$$

$$\text{Current Yield} = \text{Annual Coupon} / \text{Price}$$

$$\text{Duration } \Delta P/P \approx -\text{ModDuration} \times \Delta y$$

$$\text{Modified Duration} = \text{Macaulay Duration} / (1 + y/m)$$

$$\text{DV01} = \text{Modified Duration} \times \text{Price} \times 0.0001$$

$$\text{Yield Spread} = \text{Bond Yield} - \text{Treasury Yield}$$

$$\text{Real Yield} = \text{Nominal Yield} - \text{Inflation}$$

$$\text{Z-spread} = \text{Constant spread over entire swap curve}$$

OPTIONS

Call Intrinsic Value = $\max(S - K, 0)$

Put Intrinsic Value = $\max(K - S, 0)$

Put-Call Parity: $C - P = S - K \times e^{(-rT)}$

BS Call = $S \times N(d1) - K \times e^{(-rT)} \times N(d2)$

$d1 = [\ln(S/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T})$

$d2 = d1 - \sigma\sqrt{T}$

Delta (Call) $\approx N(d1)$

Gamma = $N'(d1) / (S \times \sigma \times \sqrt{T})$

Theta $\approx -[S \times N'(d1) \times \sigma] / (2\sqrt{T}) - rK \times e^{(-rT)} \times N(d2)$

Vega = $S \times \sqrt{T} \times N'(d1)$

RISK & POSITION SIZING

Kelly Criterion: $f^* = (bp - q) / b$

Half Kelly: $f = 0.5 \times [(bp - q) / b]$

Position Size (% risk) = $(\text{Account} \times \text{Risk\%}) / (\text{Entry} - \text{Stop})$

ATR Position: Shares = $(\text{Account} \times 2\%) / (2 \times \text{ATR})$

Volatility Target: Size = $\text{Target_Vol} / \text{Current_Vol} \times \text{Max_Size}$

Risk-Reward = $(\text{Target} - \text{Entry}) / (\text{Entry} - \text{Stop})$

Expected Value = $(\text{Win\%} \times \text{Avg_Win}) - (\text{Loss\%} \times \text{Avg_Loss})$

Profit Factor = $\text{Gross_Profit} / \text{Gross_Loss}$

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